

#ASLI ENGINEERING

Intset

SWIPE

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SET internals : Intset

Redis supports SET data structure
and it is implemented using a Hash Table
But if the set only contains integer values,
it is implemented using intset

{ "a", "b", ... }
↑
hash table

{ 1, 2, 10, 5, 3 }
↑
intset

↓

much optimized !!

Intsets are stored as a sorted list of integers

* But only until a certain limit

after the configured limit the SET
starts using HashTables to hold the data.

set-max-intset-entries

↳ does reallocation and repopulation

```
SADD k1000 1
SADD k1000 3
SADD k1000 5
```

Here we get the info about underlying encoding used to persist the key/set.

```
DEBUG OBJECT k1000
```

Complexities : Insert and Delete: O(n)

↳ But it is memory optimized

Get is O(logn)

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layout

encoding	length	int 1	int 2	.....

Because each int is fixed width 16, 32 or 64,
lookups are simple Binary Search

Adding to intset

1. checks if value can be added to intset
   eg: inserting 32 bit in 16 bit intset
2. if not: upgrade the encoding
3. do binary search & see if int exists
   if yes: do nothing and return
4. get position of insertion
   ↳ move elements using memmove
5. set int value at the position

Realloc, move, and then insert.



Deleting from intset

1. Find position of delete
2. memmove and shift elements to the left

Note: Because intset is just sorted int array,
set operations are hyper optimal.

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